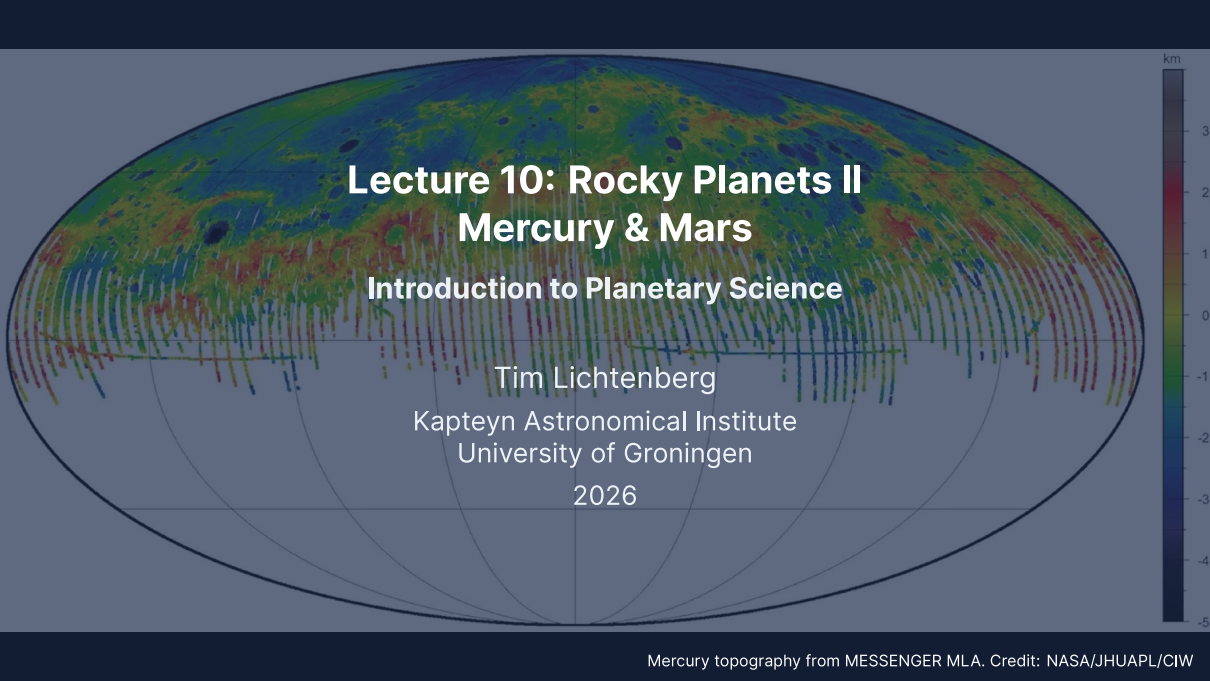


A global map of Mercury's topography in a pseudocylindrical projection. The map uses a color scale from dark blue (-5 km) to dark red (3 km). The equatorial region is predominantly red and orange, indicating higher elevations, while the polar regions are mostly blue and green, indicating lower elevations. Numerous impact craters of various sizes are visible across the surface. A vertical color bar on the right side of the map provides the elevation scale in kilometers.

Lecture 10: Rocky Planets II

Mercury & Mars

Introduction to Planetary Science

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2026

Learning Objectives

By the end of this lecture, you will be able to:

- Explain Mercury's anomalously large iron core and the spin-orbit resonance
- Describe Mars' interior, surface, and the InSight seismic results
- Derive the Jeans escape flux and apply it to atomic H, H₂, and CO₂ on Mars
- Distinguish thermal from non-thermal atmospheric escape mechanisms
- Compare Mercury and Mars as opposite limits of terrestrial-planet evolution
- Outline the missions returning data from both worlds in the 2020s and 2030s

Recap: Lecture 9

- Earth and Venus: nearly identical bulk parameters, drastically different evolution
- Simpson–Nakajima limit: maximum OLR of a saturated water atmosphere $\sim 280 \text{ W m}^{-2}$
- Venus inside the runaway-greenhouse threshold; D/H $\sim 150\times$ Earth's
- Plate tectonics + magnetic field + biosphere = the Earth-only four-way coincidence

Today: the smaller siblings, Mercury and Mars. Two opposite limits: largest iron-core fraction (Mercury) vs. near-extinct dynamo (Mars). **Why both?**



1. Mercury Overview and Surface

Caloris Basin, Mercury. MESSENGER (NASA/JHUAPL)

Mercury at a Glance

Parameter	Value	Note
Mass	$0.0553 M_{\oplus}$	smallest planet
Radius (mean)	$0.383 R_{\oplus}$	2440 km
Mean density	5429 kg m^{-3}	uncompressed $\sim 5300 \text{ kg m}^{-3}$
Semimajor axis	0.387 AU	innermost
Orbital period	87.97 d	
Rotation period	58.65 d	3:2 spin-orbit resonance
Eccentricity	0.206	largest of any planet
T_{surf} (day/night)	700/100 K	no atmosphere to redistribute
Magnetic field	$\sim 1\%$ Earth's	weak but global

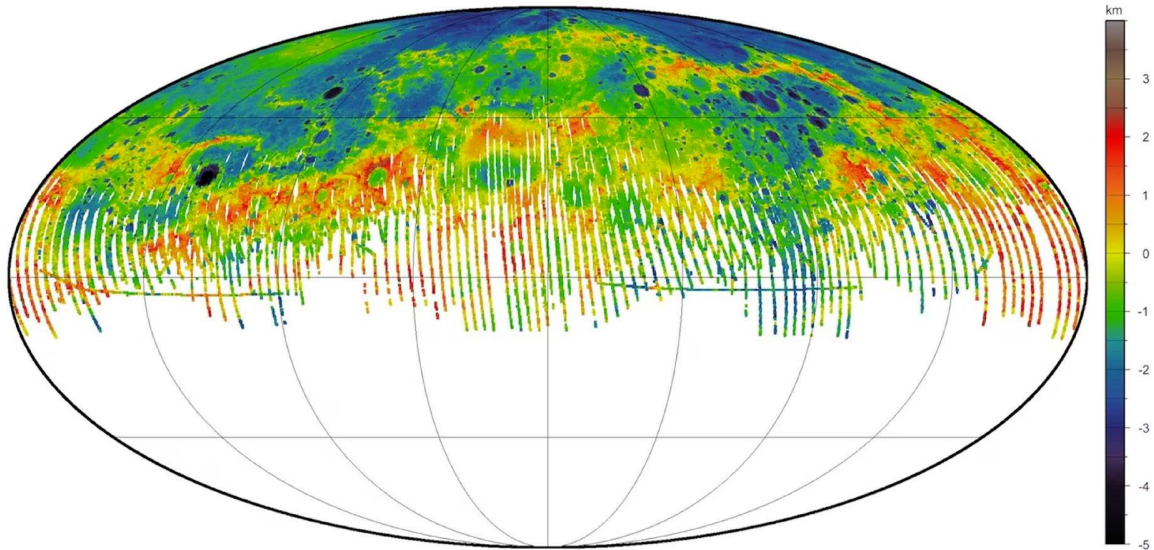
Source: NASA Mercury Fact Sheet (NSSDC)

The 3:2 Spin-Orbit Resonance

Mercury rotates exactly 3 times for every 2 orbits.

- Discovered by radar, Pettengill & Dyce (1965)
- Solar tidal torque on a permanent quadrupole moment locks Mercury into the resonance
- Stable because of high orbital eccentricity ($e = 0.206$)
- Solar day on Mercury: 176 Earth days = 2 orbits
- Hot pole and cold pole alternate with orbital phase

Mercury is the only known planet in a non-1:1 spin-orbit resonance. The capture is dynamically robust because of eccentricity-driven tidal evolution.



MESSENGER MLA global topography. Credit: NASA/JHUAPL/CIW

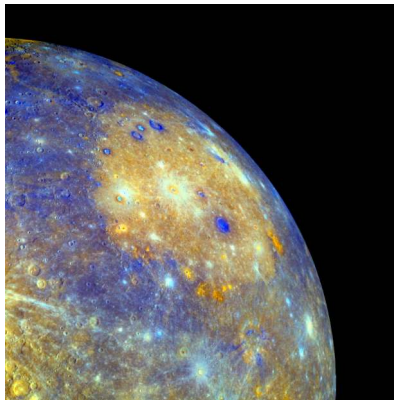
Reading Mercury's Topography

MESSENGER's Mercury Laser Altimeter mapped the planet at ~ 250 m vertical resolution.

- Total topographic range: ~ 10 km (less than Mars or the Moon)
- Highlands: ancient cratered terrain, > 4 Gyr old
- Smooth plains: volcanic infill, ~ 3.7–3.9 Ga
- **Caloris Basin** (~ 1550 km diameter): one of the largest impacts in the inner solar system
- **Lobate scarps**: thrust faults from global contraction

Two geological epochs: pre-tectonic + heavy bombardment, then a long stagnant cooling.

Caloris Basin: A Mercury-Defining Impact



Credit: NASA/JHUAPL (MESSENGER)

- Diameter ~ 1550 km, formed ~ 3.8 Ga
- Antipodal "weird terrain" from focused seismic shockwave
- Concentric ring structure preserved
- Floor flooded by basaltic plains
- Major chronostratigraphic reference for Mercury

Lobate Scarps: Mercury Has Shrunk

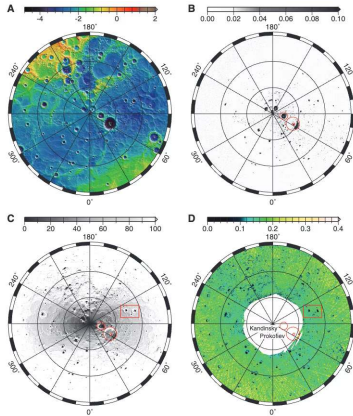


Credit: NASA/JHUAPL (MESSENGER)

- Mercury contracted as it cooled: total radial shrinkage ~ 7 km
- Compressional thrust faults form lobate scarps
- Scarp height up to 3 km, length > 1000 km
- Sets a constraint on heat-flow history

Source: Byrne et al. (2014)

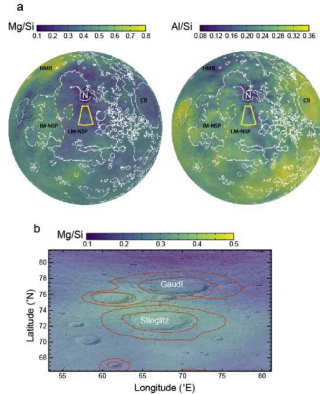
Polar Ice on the Hottest Planet



Credit: Neumann et al. (2013)

- Permanently shadowed craters at the poles never see sunlight
- Local $T < 100$ K: cold trap for water ice
- MLA reflectance + neutron spectrometer + biaxial illumination together pin down identity as H_2O ice
- Source: cometary or asteroid impacts

Mercury's Surface Chemistry



Credit: Nittler et al. (2020)

Surprises from MESSENGER X-ray and gamma-ray spectrometers:

- High volatile abundances (S, K, Cl, Na) despite 700 K daytime
- Low FeO (< 2%): mantle is highly reducing
- Mg/Si ratio higher than Earth's
- Constrains formation under low oxygen fugacity

Exosphere and Magnetosphere

Mercury has no real atmosphere; instead, a tenuous **exosphere** of atoms knocked off the surface.

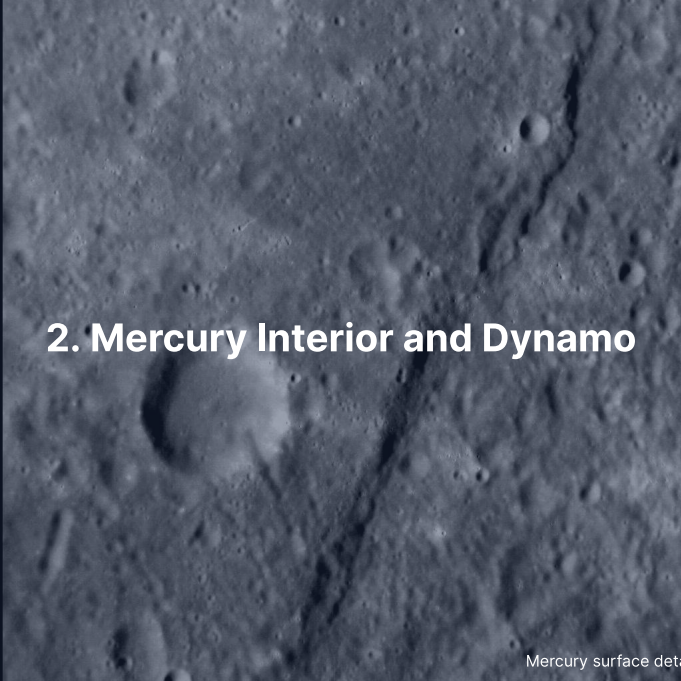
- Surface pressure $\sim 10^{-12}$ bar (effectively vacuum)
- Species detected: Na, K, Ca, Mg, He, H
- Sources: micrometeorite impact, photon-stimulated desorption, sputtering, thermal release
- Sodium tail extends $\sim 10^6$ km anti-sunward

Magnetosphere: small, asymmetric, driven by the planet's weak global dipole + solar-wind interaction.

Missions to Mercury

- **Mariner 10** (1974–1975): three flybys, ~ 45% of surface imaged
- **MESSENGER** (2011–2015): orbiter, full surface mapping, MLA, MAG, X/GRS
- **BepiColombo** (ESA/JAXA, launched 2018): two orbiters, Mercury orbit insertion late 2026
 - MPO: surface, interior, exosphere
 - Mio (MMO): magnetosphere, plasma environment

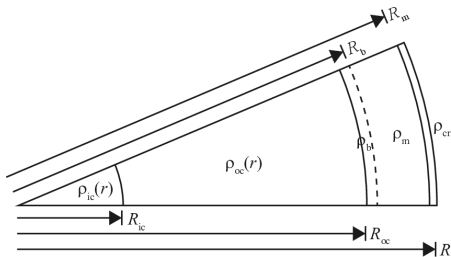
In a year and a half, BepiColombo's two orbiters will revolutionise our view of Mercury's interior, magnetosphere, and surface composition.

A grayscale image of the Mercury surface, showing a dense field of craters of various sizes. A prominent, dark, linear feature, likely a fault or a deep crater, runs diagonally from the upper right towards the lower center. The surface texture is highly irregular due to the numerous impact craters.

2. Mercury Interior and Dynamo

Mercury surface detail. MESSENGER (NASA/JHUAPL)

A Giant Iron Core

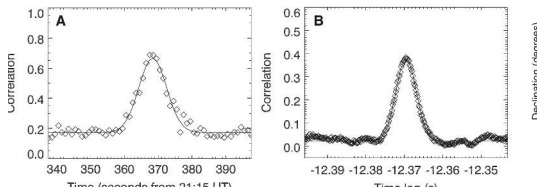


Credit: Margot et al. (2018)

Margot et al. libration measurements pin down the layers:

- Crust: ~ 35 km
- Mantle (silicate): ~ 400 km
- Outer core (liquid): below $\sim 2440 - 400 - 35 = 2005$ km radius
- Inner core (solid, possibly Fe-S)
- Core radius ~ 2020 km, $\sim 83\%$ of planetary radius
- Core mass fraction $\sim 65\%$

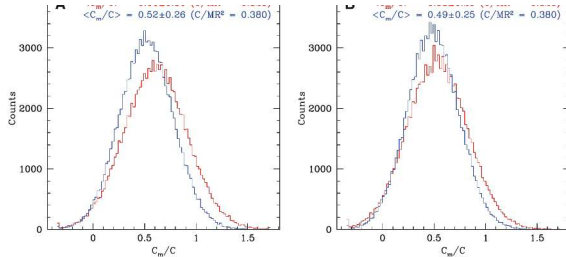
How We Know: Forced Libration



Credit: Margot et al. (2007)

- Mercury rocks slightly back and forth as it orbits
- Libration amplitude depends on whether the mantle is decoupled from the core
- Earth-based radar speckle interferometry: amplitude $\sim 0.6'$
- Implies a liquid outer core: only the silicate mantle librates freely

The Libration Measurement



Credit: Margot et al. (2007)

- Phase shift between observed and rigid-body libration
- Direct evidence the silicate mantle floats on a fluid core
- Combined with C_{20} and C_{22} from gravity, fully constrains C/MR^2 and core radius

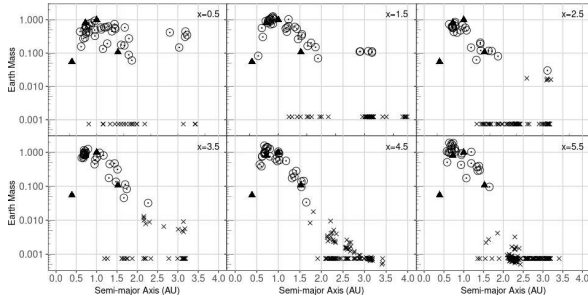
Why is Mercury so Iron-Rich?

Three competing hypotheses for Mercury's anomalous core fraction:

1. **Mantle-stripping giant impact** (Benz, Slattery & Cameron 1988): one or more late hit-and-run collisions removed silicate
2. **Photophoretic / magnetic sorting** in the protoplanetary disk: iron grains preferentially retained near the Sun
3. **Vapour fractionation**: high temperature evaporated silicates while iron remained

Compositional constraints (volatiles, low FeO) increasingly favour models without extreme heating.

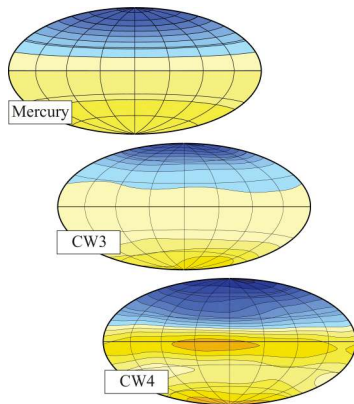
Impact-Stripping Outcomes



Credit: Franco et al. (2022)

- SPH simulations: head-on vs grazing impacts
- Mantle-stripping efficiency depends on impact angle and velocity
- Multiple hit-and-run encounters can deplete mantle without melting
- Modern Mercury composition consistent with a few high-velocity grazing impacts

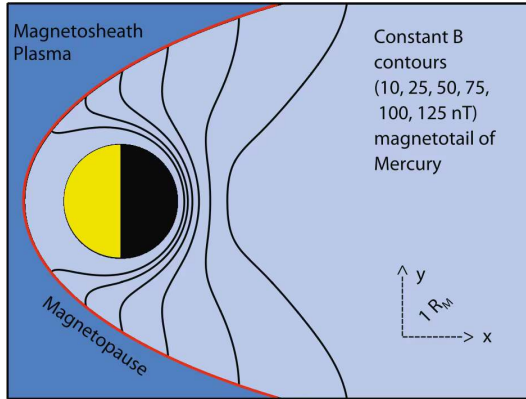
A Weak, Offset Dynamo



Credit: Wicht & Heyner (2014)

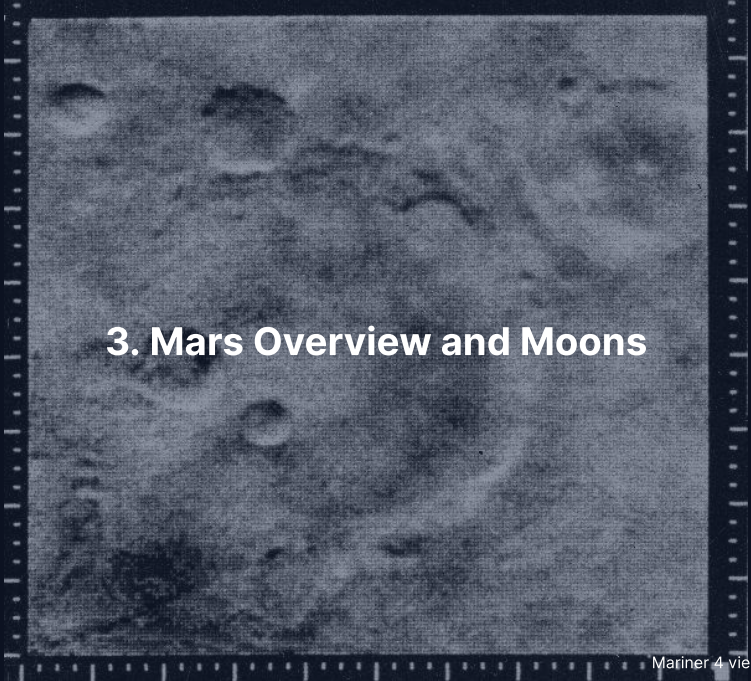
- Mercury's dipole offset $\sim 0.2 R_{Hg}$ northward
- Surface field strength: $\sim 200\text{--}400$ nT (Earth's: $\sim 50,000$ nT)
- Generated in a thin convecting outer-core shell
- Iron-snow dynamics from the bottom-up may modulate the field

Mercury's Magnetosphere



Credit: Wicht & Heyner (2014)

- Smallest planetary magnetosphere in the solar system
- Subsolar magnetopause: $\sim 1.5 R_{Hg}$
- Reconnection events drive intense flux-transfer events
- BepiColombo will measure plasma + field structure

A grainy, black and white photograph of the Martian surface, showing numerous craters of various sizes. The image is framed by a dark border with white tick marks, resembling a film strip. The text "3. Mars Overview and Moons" is overlaid in white, bold font in the center of the image.

3. Mars Overview and Moons

Mariner 4 view of Mars (1965). NASA/JPL

Mars at a Glance

Parameter	Value	Note
Mass	$0.107 M_{\oplus}$	one tenth of Earth
Radius (mean)	$0.532 R_{\oplus}$	3389 km
Mean density	3934 kg m^{-3}	lowest of the rocky planets
Semimajor axis	1.524 AU	
Orbital period	686.97 d	
Rotation period	24.62 h	similar to Earth
Axial tilt	25.2°	seasons like Earth's
Eccentricity	0.0934	strong climate forcing
T_{surf} (day/seasonal)	-130 to $+30^{\circ}\text{C}$	
Surface pressure	$\sim 6 \text{ mbar}$	0.6% of Earth's
Magnetic field	none global	local crustal anomalies

Source: NASA Mars Fact Sheet (NSSDC)

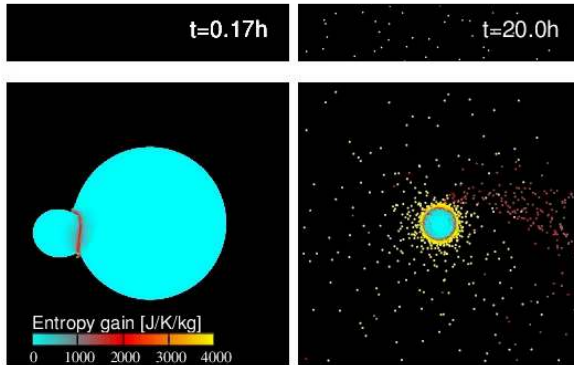
Phobos and Deimos

Mars' two tiny moons:

- **Phobos:** 22 km mean diameter, orbit at 9376 km, period 7.7 h (faster than Mars rotation)
- **Deimos:** 12 km, orbit at 23463 km, period 30.3 h
- Both have very low density ($\sim 1900 \text{ kg m}^{-3}$), consistent with rubble-pile internal structure
- Two competing origin scenarios:
 - Captured asteroids
 - In situ formation from a giant-impact debris disk

Phobos is below synchronous orbit and will tidally decay onto Mars in ~ 30 Myr.

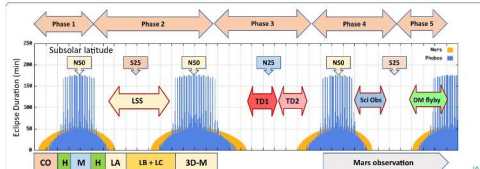
The Giant-Impact Origin Hypothesis



Credit: Hyodo et al. (2017)

- SPH simulation of a Borealis-scale impact
- Forms an extended debris disk around Mars
- Phobos and Deimos accrete from the disk
- Reproduces orbital and compositional constraints
- Predicts a fraction of Mars-derived material in the moons

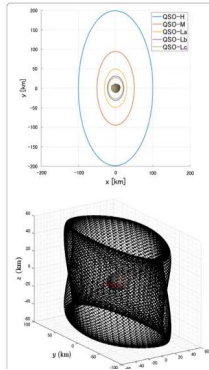
MMX: Sample Return from Phobos



Credit: JAXA

- JAXA's Martian Moons eXploration mission
- Launch 2026, sample return to Earth 2031
- Will distinguish capture vs. impact origin
- Carries a NASA gamma-ray + neutron spectrometer

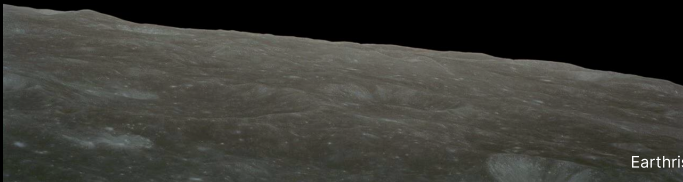
MMX Mission Architecture



Credit: JAXA

- Quasi-satellite orbit around Phobos for surface mapping
- Surface sampling phase: ~ 10 g of regolith
- Earth return capsule decouples for atmospheric entry

Break



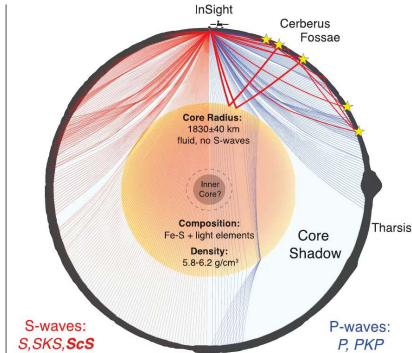
Earthrise, Apollo 8 (1968). Credit: NASA

The image displays four terrestrial planets in a row, scaled to show their relative sizes. From left to right, they are Mercury (a small, grey, cratered sphere), Venus (a large, yellowish-brown sphere with a thick, hazy atmosphere), Earth (a large sphere with blue oceans, white clouds, and brown/green landmasses), and Mars (a medium-sized sphere with a reddish-orange surface and polar ice caps). The planets are set against a solid dark blue background. The title text is centered over the Venus and Earth planets.

4. Mars' Interior: The InSight Revolution

Terrestrial planets to scale. Credit: NASA/JPL-Caltech

Mars' Interior, Resolved

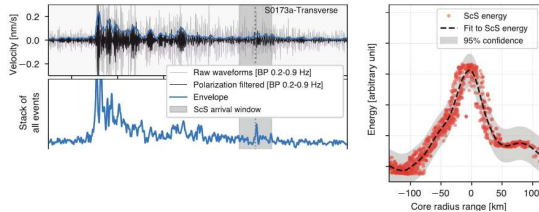


Credit: Stähler et al. (2021)

InSight SEIS results:

- Crust: 24–72 km thick, thinner in the north, thicker in the south
- Mantle to ~ 1560 km depth
- Olivine–wadsleyite transition near ~ 1100 km
- **Liquid iron-alloy core** of radius ~ 1830 km
- Core density ~ 6000 kg m^{-3} (Earth: ~ 10,900)

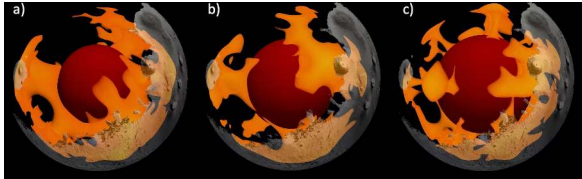
Reading a Marsquake Waveform



Credit: Stähler et al. (2021)

- P-wave arrival, then S-wave arrival
- Core reflections (ScS) absent: outer core is liquid
- ~ 1300 events catalogued during the mission
- Largest event: S1222a, $M = 4.7$, May 2022

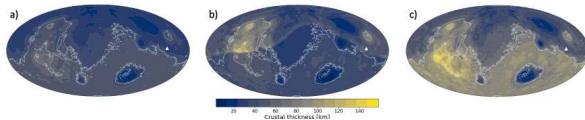
Mars Mantle Convection Today



Credit: Pleša et al. (2022)

- 3D thermal-evolution model consistent with InSight constraints
- Single-plume convection regime
- Sluggish lower-mantle flow
- Tharsis upwelling persists to the present

Mars Crustal Thickness Map



Credit: Pleša et al. (2022)

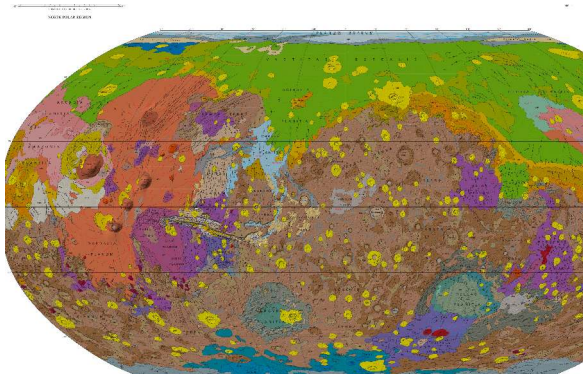
- Northern lowlands: thin crust, ~ 25–35 km
- Southern highlands: thick crust, ~ 50–70 km
- Hemispheric dichotomy (likely Borealis impact origin)
- Sets the long-term cooling pattern



5. Mars Geology and Surface Highlights

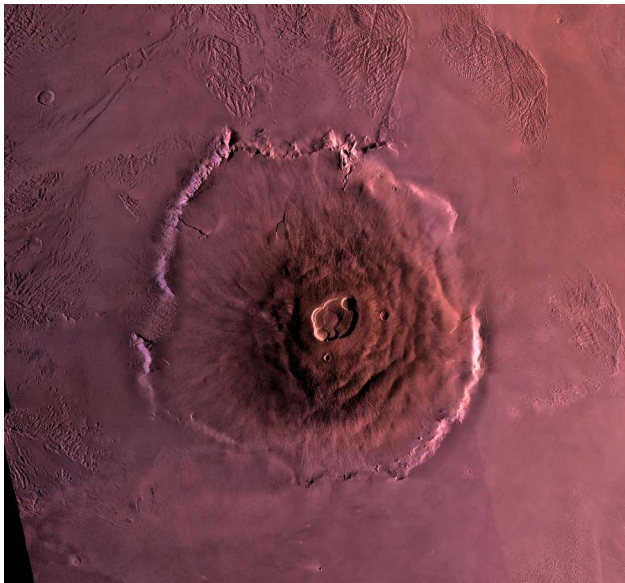
Olympus Mons. NASA/JPL/Mars Express (ESA)

Mars Geological Map



Credit: Tanaka et al. (2014)

- Tharsis volcanic province dominates the western hemisphere
- Hellas, Argyre, Isidis: large impact basins
- Northern lowlands: relatively young, partly buried
- Map gives direct visual of stratigraphic ages

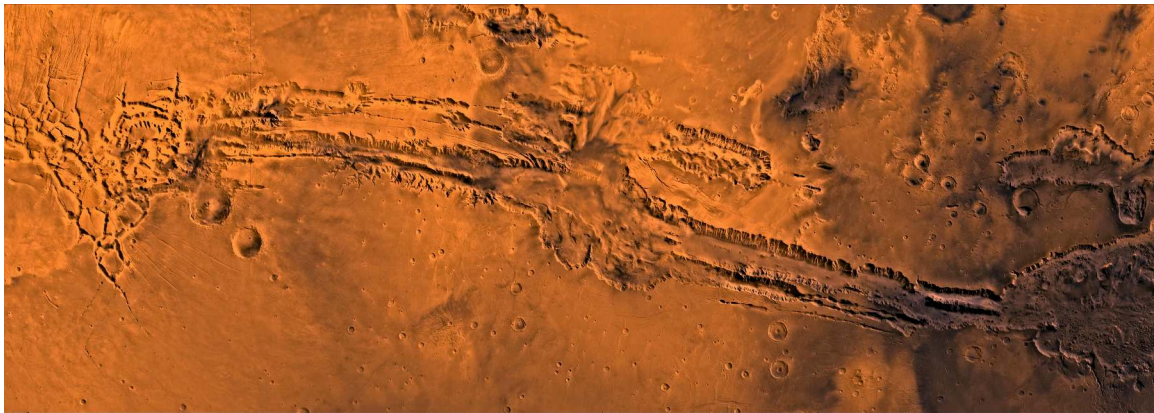


Olympus Mons, Mars Express. Credit: ESA/DLR/FU Berlin

Olympus Mons: The Tallest Volcano

- Height: 21.9 km above the reference geoid (Earth's tallest, Mauna Kea, is 10 km above the sea floor)
- Diameter: 600 km (the size of Italy)
- Shield volcano with low slope angles
- Calderas at the summit: ~ 80 km across
- Flank scarps: ~ 6 km cliffs
- Stagnant-lid plumes can pile up to enormous heights without lateral plate motion to spread the load

Tharsis as a whole is loaded enough to have plausibly reoriented Mars' rotation axis ("true polar wander").



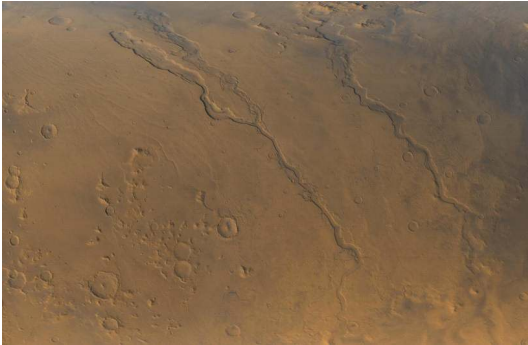
Valles Marineris, Viking mosaic. Credit: NASA/JPL-Caltech

Valles Marineris: A Tectonic Canyon

- 4000 km long, 200 km wide, up to 7 km deep
- As wide as the United States; as deep as four Grand Canyons stacked
- Origin: extensional rifting linked to Tharsis loading + uplift
- Outflow channels exit the eastern end into chaotic terrain
- Possibly lake-filled in the Hesperian; layered sediments observed

Valles Marineris is not an erosional canyon; it is a tectonic graben modified by collapse and subsequent fluvial activity.

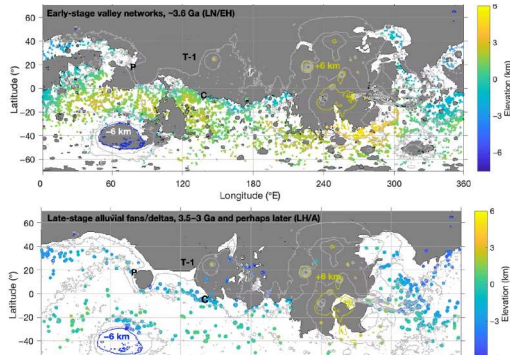
Past Water: Valley Networks



Credit: NASA/JPL/USGS

- Dendritic drainage networks on Noachian terrain
- Confluences, meanders, tributaries: liquid-water erosion
- Densities comparable to Earth's drainage basins
- Active for > 100 Myr in the late Noachian

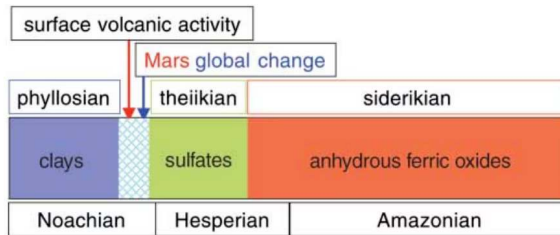
Mapping the Wet Era



Credit: Kite et al. (2022)

- Latitudinal distribution of valley networks
- Concentrated at low to mid-latitudes
- Pattern matches a sustained climate (not single-event flooding)
- Constrains the duration of habitable surface conditions

Phyllosian, Theiikian, Siderikan: Mineralogy Tells Time

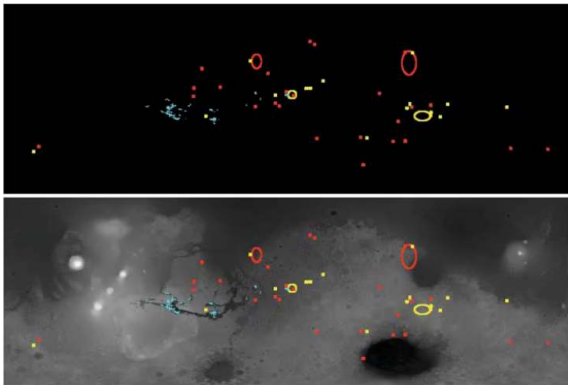


Credit: Bibring et al. (2006)

OMEGA (Mars Express)
mineralogy framework:

- **Phyllosian** (> 4 Ga): clays, neutral-pH water
- **Theiikian** (3.7–3.5 Ga): sulphates, acidic water
- **Siderikan** (< 3 Ga): hematite, oxidising, dry

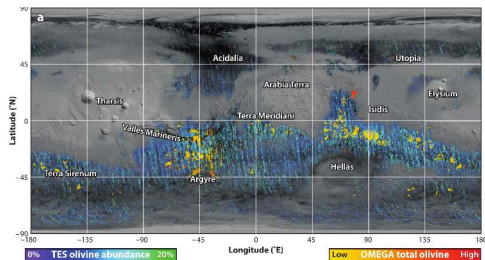
Phyllosilicate and Sulphate Distributions



Credit: Bibring et al. (2006)

- Red: phyllosilicates (clays)
- Blue: sulphates
- Yellow: other hydrated minerals
- Spatial pattern ties to age via the timeline above

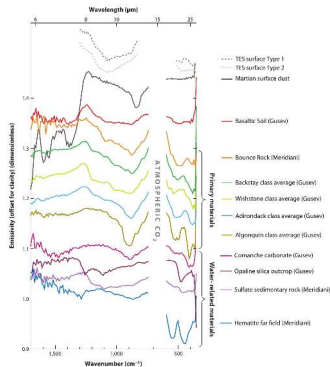
Olivine: The Aqueous Alteration Boundary



Credit: Ehlmann & Edwards (2014)

- Olivine is unstable in liquid water (alters to serpentine + clays)
- Detection of fresh olivine = surfaces never sustainedly wet
- Olivine common across the equatorial region: most of Mars never had a long surface ocean

CRISM Spectra: How We See the Minerals



Credit: Ehlmann & Edwards (2014)

- Diagnostic absorption features in the near-IR
- Phyllosilicates: 1.4, 1.9, 2.2 μm
- Sulphates: 1.4, 1.9, 2.4 μm
- Olivine: broad band at 1 μm

6. Blackboard Derivation: Jeans Escape Flux



Setup: The Exobase

Goal: derive the rate at which thermal energy alone allows molecules to escape from a planetary atmosphere; apply it to Mars.

The exobase. Altitude where the mean free path equals the atmospheric scale height. Above it, molecules move ballistically.

Most-probable speed in Maxwell-Boltzmann distribution:

$$v_{\text{th}} = \sqrt{2k_{\text{B}}T/m}$$

Escape velocity at the exobase:

$$v_{\text{esc}} = \sqrt{2GM/r_{\text{exo}}}$$

Step 1: The Escape Parameter

Define dimensionless ratio:

$$\lambda \equiv \left(\frac{v_{\text{esc}}}{v_{\text{th}}} \right)^2 = \frac{G M m}{k_{\text{B}} T r_{\text{exo}}}$$

- $\lambda \gg 1$: molecules gravitationally bound; escape rare
- $\lambda \lesssim 1$: thermal energy comparable to escape energy; rapid loss
- Depends strongly on molecular mass m

Step 2: Maxwell-Boltzmann Distribution

At the exobase, gas in local thermal equilibrium:

$$f(v) dv = n \left(\frac{m}{2\pi k_B T} \right)^{3/2} 4\pi v^2 \exp\left(-\frac{mv^2}{2k_B T}\right) dv$$

Number flux of upward-moving molecules with $v_z > 0$:

$$\Phi = \int_{\text{up}} v_z f(\mathbf{v}) d^3v$$

We need v_z such that the speed exceeds v_{esc} .

Step 3: The Jeans Formula

Integrating over the upper hemisphere of velocity space, with the speed cutoff $v \geq v_{\text{esc}}$:

$$\Phi_J = n \langle v \rangle \frac{1}{4} (1 + \lambda) e^{-\lambda}$$

where $\langle v \rangle = \sqrt{8k_B T / (\pi m)}$ is the mean speed.

Substituting and simplifying:

$$\Phi_J = n \sqrt{\frac{k_B T}{2\pi m}} (1 + \lambda) e^{-\lambda}$$

This is the Jeans escape flux (Jeans 1925).

Step 4: Checking the Limits

- $\lambda \rightarrow 0$: $(1 + \lambda)e^{-\lambda} \rightarrow 1$, formula reduces to standard kinetic-theory effusion flux $n\sqrt{k_B T / (2\pi m)}$
- $\lambda \gg 1$: exponential decay dominates; flux $\propto \lambda e^{-\lambda}$
- Steepness: a change in λ from 5 to 10 reduces escape by a factor of ~ 80

Jeans escape is exponentially sensitive to λ : small changes in temperature or molecular mass change the loss rate by orders of magnitude.

Step 5: Apply to Mars

$$M = 6.4 \times 10^{23} \text{ kg}, r_{\text{exo}} \approx 3.6 \times 10^6 \text{ m}, T_{\text{exo}} \approx 270 \text{ K}.$$

Escape velocity at the exobase:

$$v_{\text{esc}} \approx 4.9 \text{ km/s}$$

Species	λ	$e^{-\lambda}$
H	5.3	5×10^{-3}
H ₂	10.6	5×10^{-5}
He	21.2	$\sim 10^{-9}$
O	84.8	10^{-37}
CO ₂	230	10^{-100}

Mars cannot lose CO₂ thermally. Only H and H₂ escape via Jeans; heavier species need non-thermal channels.

Why Jeans Escape is the Foundation

- Underpins atmospheric escape theory in general
- Always present at any temperature above absolute zero
- Sets the lower bound on escape rate
- Independent of magnetic field, ionisation state, or solar wind

Other mechanisms (photochemical, sputtering, ion pickup, hydrodynamic) operate **in addition** to Jeans, with different mass selectivities and intensities.

An aerial photograph of the Martian surface, showing a complex network of dry river valleys and numerous impact craters of various sizes. The terrain is reddish-brown and appears desolate. The valleys are prominent, winding across the landscape, and the craters are scattered throughout, some with distinct rims and shadows.

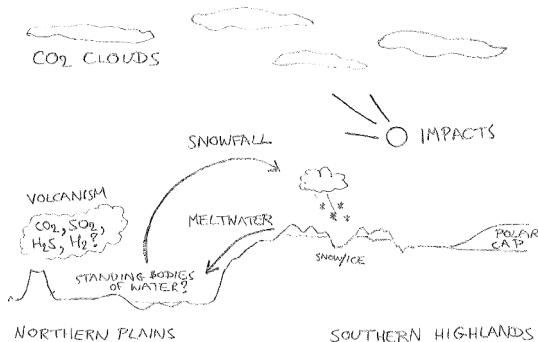
7. Mars Atmosphere and Climate Loss

The Early Mars Climate Puzzle

- Geological evidence: liquid water flowed in the late Noachian
- Solar luminosity at 4 Ga was ~ 75% of today
- Pure CO₂ greenhouse cannot warm Mars above freezing at 1.5 AU
- Required: extra greenhouse gas (CH₄, H₂ collision-induced absorption)

Active research: episodic warming + impacts + volcanism + reducing atmospheres in interplay.

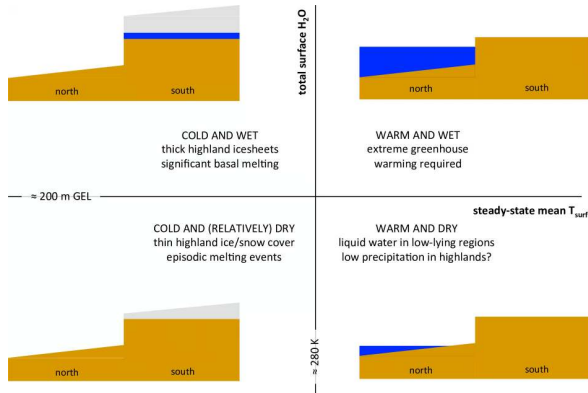
Reducing-Atmosphere Greenhouse



Credit: Wordsworth (2016)

- $\text{H}_2 + \text{CO}_2$ collision-induced absorption opens long-wavelength greenhouse
- Sustained H_2 supply requires reducing volcanism / serpentinisation
- Could keep early Mars above freezing at 1.5 AU

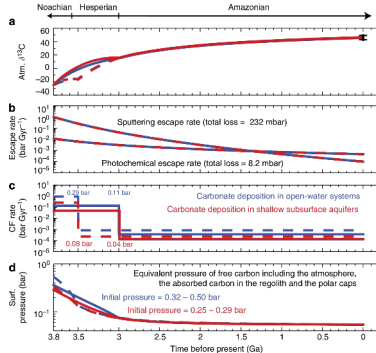
Phase Diagram: Wet vs Dry Mars



Credit: Wordsworth (2016)

- Climate state vs CO₂ pressure and volcanic H₂ flux
- Multiple equilibria possible
- Defines the early Mars habitability envelope

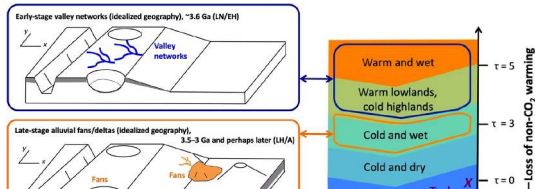
The Carbon Inventory



Credit: Hu et al. (2015)

- Coupled atmosphere-surface-interior carbon model
- Initial CO₂ inventory: 1–3 bar
- Loss split between escape, carbonate burial, polar caps
- Modern 6 mbar leaves no doubt: most of the original atmosphere is gone

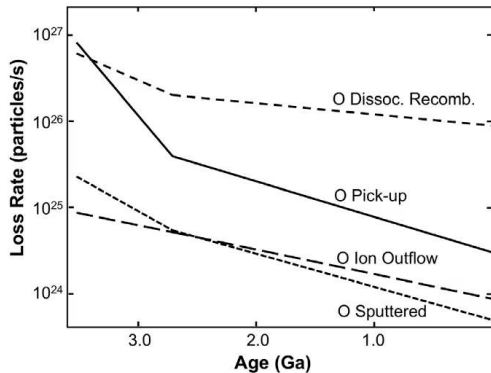
Climate Stability Mechanism



Credit: Kite et al. (2022)

- Episodic climate warming via volcanic H_2 pulses
- Linked to impacts and crustal magmatism
- Total time above freezing: ~ 100 Myr cumulative

MAVEN: Measuring Atmospheric Loss

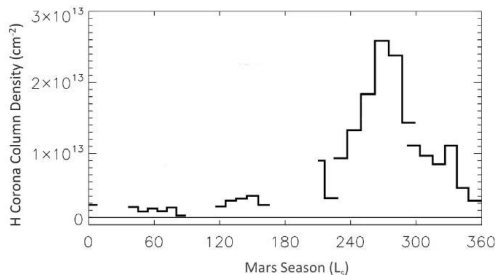


Credit: Jakosky et al. (2018)

Total escape rate ($\sim 2 \text{ kg s}^{-1}$)
split across:

- **Photochemical** O escape (dominant for O today)
- **Ion pickup** (CO_2^+ , O^+)
- **Sputtering** by solar-wind ions
- **Polar plume** non-thermal H loss

Hydrogen Loss Channels



Credit: Jakosky et al. (2018)

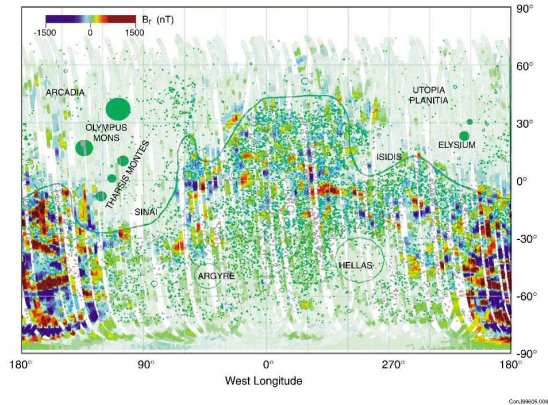
- H is lost via Jeans + non-thermal mechanisms
- Escape rate varies by 1–2 orders of magnitude over a Mars year
- Strongly modulated by solar activity
- Implies a much wetter Mars in the deep past



8. Mars Magnetism and Mission History

Early Mars from Mariner 4 (1965). Credit: NASA/JPL

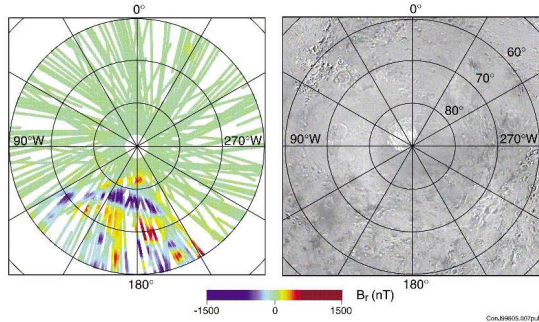
Mars Crustal Magnetism



Credit: Acuña et al. (1999)

- No global field today
- Strong remnant magnetisation in the southern highlands
- Implies a vigorous early dynamo, then shutoff
- Northern lowlands: weak (impact-erased or younger crust)

Lineated Magnetic Anomalies



Credit: Acuña et al. (1999)

- Stripe-like patterns suggestive of polarity reversals
- Recorded by basaltic crust during cooling
- Pre-Borealis-impact crust preserves the early dynamo signature
- Extinction of the dynamo: ~ 4.1 Ga

Mars Mission History (Selected Highlights)

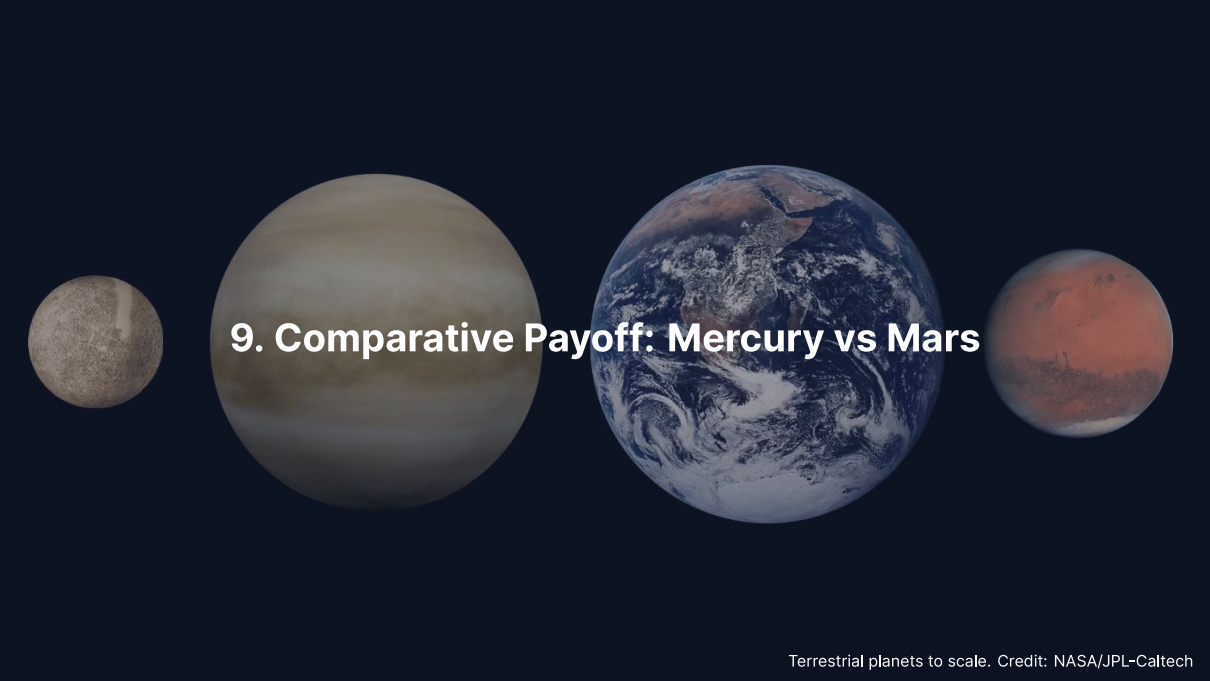
- **Mariner 4** (1965): first flyby; cratered "Moon-like" Mars
- **Viking 1&2** (1976): first landers, first life-detection experiments (ambiguous results)
- **MGS, Odyssey, MEX, MRO, Maven**: orbital era 1996–present
- **Spirit, Opportunity, Curiosity, Perseverance**: rovers
- **InSight** (2018–2022): seismometer + heat-flow probe
- **Tianwen-1** (2021, China): orbiter + Zhurong rover
- **Mars Sample Return**: planned 2030s

Mars Sample Return: The Next Decade

Perseverance (since 2021) is caching samples from Jezero Crater, an ancient lake bed.

- Cores: igneous, sedimentary, organic-bearing rocks
- ESA + NASA architecture under revision (cost overruns)
- Sample return earliest in the early 2030s
- Will allow precise radiometric dating + biosignature analysis

Sample return is the difference between "consistent with biological origin" and "biological origin proven beyond reasonable doubt".

The image displays four terrestrial planets in a row, scaled to show their relative sizes. From left to right, they are Mercury (a small, grey, cratered sphere), Venus (a large, yellowish-brown sphere with horizontal cloud bands), Earth (a large sphere with blue oceans and white clouds), and Mars (a medium-sized, reddish-brown sphere with darker patches). The text "9. Comparative Payoff: Mercury vs Mars" is centered over the planets in white.

9. Comparative Payoff: Mercury vs Mars

Mercury and Mars as Opposite Limits

Property	Mercury	Mars
Distance from Sun	0.39 AU	1.52 AU
Mass ($M_{\oplus}$)	0.055	0.107
Core fraction	~ 65%	~ 25%
Mantle thickness	~ 400 km	~ 1560 km
Surface volatiles	cold trap ice only	once-flowing water
Magnetic field today	weak global	none global
Atmosphere	exosphere	6 mbar CO ₂
Tectonics	shrinking, lobate scarps	stagnant lid

Mercury sits at the maximum-iron, no-volatile end. Mars sits at the maximum-volatile, low-iron end. Earth and Venus fall in between.

Size and Distance Set the Trajectory

- **Small bodies cool fast:** surface-area-to-volume ratio $\propto 1/R$
- Mars cools faster than Earth $\Rightarrow$ early dynamo shutoff
- Mercury's massive core preserves heat $\Rightarrow$ dynamo still active (just barely)
- **Distance from the Sun** sets atmospheric escape and irradiation history

Combined: small + warm $\Rightarrow$ Mercury (volatile-poor, iron-dominant). Small + cold $\Rightarrow$ Mars (water-rich early, escapes to space later).

The Timing Problem: Dynamo Lifetimes

- Earth: dynamo active for ≥ 3.5 Gyr, still going
- Mars: dynamo $\sim 4.5\text{--}4.1$ Ga, then off
- Mercury: dynamo weak but still active
- Venus: dynamo unclear, possibly dead

Lesson: dynamo persistence depends on (a) core size, (b) cooling rate of the mantle, (c) chemical buoyancy from inner-core nucleation. Mars failed (b); Mercury preserves (a); Earth has all three.

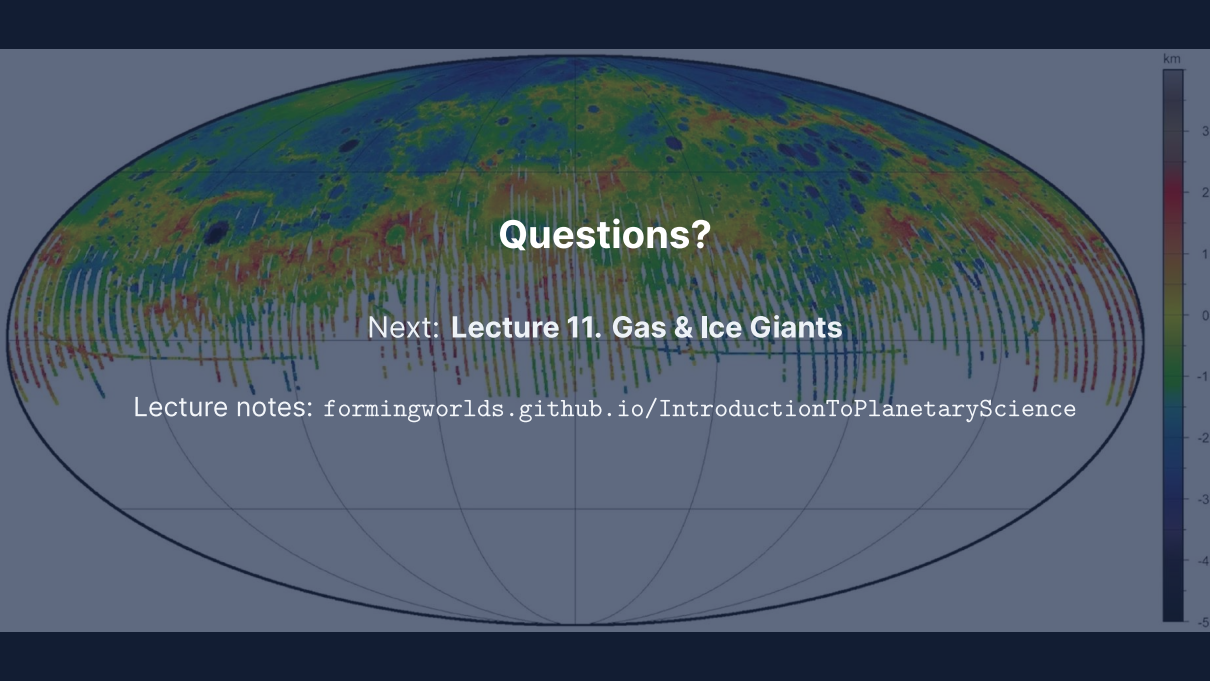
Comparative View: What Makes a Rocky Planet Habitable?

- Stellar flux inside the runaway-greenhouse limit (rules out Venus)
- Long-lasting atmosphere (rules out Mercury, late Mars)
- Liquid water reservoir at the surface (rules out Mercury entirely)
- Internal heat sufficient for tectonic recycling (rules out Mars after ~ 1 Ga)
- Magnetic shield against solar-wind erosion (helps but not strictly required)

Habitability is a multi-dimensional constraint. Each of Mercury, Venus, Mars fails on different axes; Earth happens to satisfy all of them simultaneously.

Summary

1. Mercury: 3:2 spin-orbit resonance, 65 percent iron core, weak offset dynamo, polar ice in cold traps
2. Mars: half Earth's radius, one tenth its mass, lost its dynamo at ~ 4.1 Ga and most of its atmosphere thereafter
3. Jeans escape: $\Phi_j \propto (1 + \lambda)e^{-\lambda}$, mass-selective; H and H_2 escape Mars, CO_2 does not
4. Non-thermal escape (photochemical, ion pickup, sputtering) drives the heavy-species loss
5. InSight delivered the first marsquake catalogue and confirmed a liquid Mars core
6. Mercury and Mars sit at opposite limits of terrestrial-planet evolution: maximum core fraction vs. maximum atmospheric volatile loss
7. BepiColombo (2026) and MMX (2026) will revolutionise our view of both planets in the next few years



Questions?

Next: **Lecture 11. Gas & Ice Giants**

Lecture notes: formingworlds.github.io/IntroductionToPlanetaryScience